

1) Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

RAM Reseating – Step-by-Step Documentation

1. **Shut down the computer** and disconnect the power cable.
2. Press the power button for a few seconds to discharge any remaining power.
3. **Open the computer case** and locate the RAM slots on the motherboard near the CPU socket.
4. Identify the installed RAM module and note its position.
5. Open the locking clips at both ends of the RAM slot.
6. Hold the RAM module by its **edges** and carefully remove it from the slot. Avoid touching the gold contacts.
7. Inspect the RAM module and slot for dust, damage, or bent contacts.
8. Align the notch on the RAM module with the key in the RAM slot.
9. Press the RAM module firmly and evenly into the slot until the locking clips click into place.
10. Reconnect the power cable and switch on the computer.
11. Check whether the computer successfully completes **POST** and displays the normal boot screen.
12. Enter BIOS/UEFI if necessary and verify that the correct amount of RAM is detected.
13. Close the computer case after confirming that the system is working correctly.

Issues Observed

- RAM module was successfully removed and re-inserted.
- No physical damage or bent contacts were observed.
- RAM was securely locked into the slot after reseating.
- The computer completed POST normally.
- The installed RAM capacity was detected correctly in BIOS/UEFI.
- **No issues were observed during the reseating process.**

2) Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.
Hint: Think about display issues, driver checks, and physical connections.

1. Identify the Symptoms

Check for no display, black screen, screen flickering, graphical artifacts, crashes, or poor gaming performance.

Note when the problem occurs—for example, during startup, gaming, or after installing software.

2. Check the Display Connection

Make sure the monitor is powered on and set to the correct input source.

Check the HDMI/DisplayPort cable and try another cable or monitor if available.

Ensure the display cable is connected to the GPU output, not the motherboard video output when using a dedicated GPU.

3. Check the Physical GPU Installation

Shut down and unplug the PC.

Confirm that the GPU is firmly seated in the PCIe slot.

Check that the GPU's PCIe power connectors are securely connected.

Inspect for dust, loose connections, or visible physical damage.

4. Check GPU Drivers

Start Windows and open Device Manager.

Check Display adapters for warning symbols or errors.

Update or reinstall the correct GPU driver from the manufacturer's official website.

Restart the PC after installing the driver.

5. Check GPU Temperature and Performance

Monitor GPU temperature while idle and during gaming.

Check whether the GPU fans are operating normally.

Look for overheating, sudden performance drops, crashes, or artifacts.

3) Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting

Storage Drive Disconnect/Reconnect Test

Test system: Desktop PC with SATA SSD/HDD

Purpose: To observe how the system behaves when the storage drive is temporarily disconnected.

1. Power off the PC and disconnect the power cable.
2. Open the case and locate the SATA data and SATA power connectors connected to the SSD/HDD.
3. Disconnect the SATA data cable (or power cable) from the drive.
4. Reconnect the power and start the PC.
5. Before reconnecting the drive, the BIOS/UEFI may show:
 - The drive missing from the Storage/SATA Information section.
 - A message such as “No bootable device found”, “Boot device not found”, or “No operating system found.”
 - The system may enter the BIOS automatically because no boot drive is detected.
6. Shut down the PC again and reconnect the SATA cable securely.
7. Start the PC and enter BIOS/UEFI.
8. After reconnecting the drive, confirm that the SSD/HDD model and storage capacity appear in the SATA/Storage Information section.
9. Allow the computer to boot into Windows.
10. Open Disk Management or File Explorer and verify that the storage drive is detected correctly.
11. Confirm the fix by restarting the PC and checking that it boots normally.

4) Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

Diagnostic Steps for RAM

1. Power off the PC and disconnect it from the mains.
2. Open the case and check whether the RAM modules are properly seated in their slots.
3. Remove and carefully reseal the RAM, making sure both locking clips are fully engaged.
4. Check the RAM and slots for dust, physical damage, or corrosion.
5. If multiple RAM modules are installed, test them one at a time to identify a potentially faulty module.

6. Run Windows Memory Diagnostic or MemTest86 to check for memory errors.
7. Check the BIOS/UEFI to confirm that the correct RAM capacity and speed are detected.
8. Run the PC under normal or heavy workload and monitor whether the random restarts continue.
9. If memory errors are detected, replace the faulty RAM module with a compatible known-good module.
10. Confirm the fix by running a memory test again and operating the PC for an extended period without unexpected restarts.